

SAN MARTIN, Argentina

WMO: 874160

Lat: 33.0733S

Lon: 68.4650W

Elev: 2142

StdP: 13.59

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.9	33.5	16.4	13.6	46.2	19.6	15.9	45.4	20.6	53.2	16.6	49.5	1.8	230	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.5	95.1	69.5	92.6	69.0	90.2	68.3	73.8	89.1	72.1	87.4	70.7	85.5	5.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	115.7	82.3	67.3	108.6	80.8	65.6	102.5	79.2	39.0	88.9	37.4	87.5	36.0	85.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.6	16.1	14.7	DB	27.0	100.9	2.4	2.8	25.3	102.9	23.9	104.6	22.5	106.2	20.8	108.2
			WB	24.6	77.6	2.2	2.1	23.1	79.1	21.8	80.3	20.6	81.5	19.0	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	78.5	75.7	70.5	62.4	55.1	49.2	48.0	52.8	58.6	66.0	71.8	76.5	(6)
	DBStd	12.11	5.37	5.96	5.98	6.20	5.67	4.65	5.48	6.64	7.05	6.76	6.31	5.88	(7)
	HDD50	251	0	0	0	3	19	67	102	45	13	2	0	0	(8)
	HDD65	2135	1	5	20	118	307	474	527	383	210	69	17	4	(9)
	CDD50	5255	885	719	636	376	178	43	40	131	272	499	653	823	(10)
	CDD65	1664	421	304	191	41	1	0	5	19	101	219	362		(11)
	CDH74	15282	4302	2717	1347	237	20	0	1	57	185	825	1894	3697	(12)
	CDH80	6228	1998	1121	425	33	1	0	11	34	239	707	1659		(13)
(14) Wind	WSAvg	4.0	4.5	4.2	3.7	3.4	2.9	2.9	3.4	3.9	4.5	5.0	5.3	4.8	(14)
(15) Precipitation	PrecAvg	6.6	1.3	1.5	1.0	0.3	0.2	0.2	0.1	0.1	0.4	0.2	0.7	1.1	(15)
	PrecMax	11.5	4.6	4.1	3.3	1.2	1.1	1.7	0.7	0.8	1.9	1.2	2.3	3.3	(16)
	PrecMin	3.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.4	0.9	1.3	0.9	0.4	0.3	0.3	0.2	0.2	0.5	0.3	0.7	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.2	96.3	91.3	83.8	76.6	68.5	71.1	81.0	84.1	90.8	94.4	98.1	(19)
		MCWB	70.9	72.2	70.1	65.5	61.1	53.7	53.0	59.1	61.4	64.0	66.3	69.1	(20)
	2%	DB	95.3	92.6	88.0	79.8	72.2	64.8	65.5	73.5	79.0	85.8	90.6	94.5	(21)
		MCWB	70.5	70.1	69.2	63.0	58.8	51.8	50.8	55.3	58.2	62.8	65.5	68.5	(22)
	5%	DB	92.5	89.9	84.8	76.3	68.4	61.8	61.8	68.9	75.1	82.2	87.3	91.9	(23)
		MCWB	69.9	69.6	67.9	61.8	56.3	50.1	48.5	52.8	56.4	60.9	64.4	68.1	(24)
	10%	DB	89.7	87.0	81.6	72.9	64.9	58.7	58.2	64.7	71.0	78.6	84.1	88.8	(25)
		MCWB	69.1	68.7	66.5	60.4	54.5	48.5	46.5	50.4	54.4	59.3	63.3	66.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.8	75.8	74.3	69.2	63.7	56.2	55.7	61.9	64.0	68.1	70.2	74.1	(27)
		MCDB	92.8	88.9	87.3	78.9	73.6	65.3	67.7	76.5	79.4	84.6	87.1	91.3	(28)
	2%	WB	73.9	73.4	71.4	65.7	60.5	53.3	51.9	56.7	60.0	65.3	68.0	71.7	(29)
		MCDB	89.8	88.1	83.4	74.9	69.1	62.0	62.8	71.7	76.0	81.0	85.6	89.5	(30)
	5%	WB	72.1	71.7	69.5	63.6	58.1	51.3	49.4	53.6	57.5	62.6	66.2	70.0	(31)
		MCDB	88.2	85.7	81.1	73.1	66.1	59.9	60.1	67.4	72.8	78.4	83.8	87.0	(32)
	10%	WB	70.5	70.2	67.8	61.6	55.9	49.3	47.3	51.2	55.3	60.7	64.5	68.3	(33)
		MCDB	86.4	83.3	79.3	70.8	63.2	57.2	56.9	63.4	68.9	75.9	81.1	85.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.5	19.9	19.0	18.9	18.8	20.2	21.1	21.6	20.8	21.2	21.4	21.4	(35)
		MCDBR	24.4	24.2	23.8	23.8	23.4	24.4	25.4	27.2	27.0	26.4	26.4	25.9	(36)
	5% WB	MCWBR	8.9	8.3	9.7	11.5	12.6	14.4	14.5	14.6	12.9	10.9	10.3	9.6	(37)
		MCWBR	21.7	20.5	20.3	20.6	20.7	22.2	24.0	26.4	24.8	23.4	23.6	22.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.394	0.388	0.368	0.362	0.325	0.302	0.298	0.323	0.364	0.366	0.374	0.389	(40)
		taud	2.363	2.387	2.441	2.416	2.478	2.550	2.514	2.432	2.315	2.344	2.347	2.345	(41)
	Ebn at Noon		299	295	289	272	267	268	275	281	285	298	304	302	(42)
		Edh at Noon	42	39	35	32	26	23	25	30	39	40	42	43	(43)
(44) All-Sky Solar Radiation	RadAvg		2495	2177	1786	1361	983	904	989	1286	1686	2112	2484	2622	(44)
	RadStd		138	136	121	128	115	69	66	87	119	180	149	139	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+0.86	+0.74	N/A	N/A	N/A	N/A	+113

Nomenclature: See separate page